DressMe BO6

Smart Wardrobe Digitization - summary of the executed notebook and its results

Objective. Develop two ML components for a digital wardrobe: classify a garment from one photo and produce a short automatic description. The notebook audits the dataset, trains and compares classifiers, adds attribute predictions, verifies the export and implements optional photo-quality warnings.

1. Dataset audit and preparation

The Fashion Product Images dataset was used. Duplicate dataset copies were identified, and one canonical copy was selected. Metadata, per-product records and images were inspected before constructing the modelling pool.

Audit item	Recorded result
Metadata	44,446 records: 44,424 well-formed rows and 22 malformed rows repaired.
Image integrity	44,441 images fully decoded; no decoding failures.
Image / metadata matching	5 metadata records without images; no orphan images.
Exact duplicates	652 groups across the dataset; 5 mapped rows with conflicting targets excluded.
Final modelling pool	15,441 mapped garment rows with decoded images.

Taxonomy: dress, jacket, jeans, shirt, shorts, skirt, sweater, sweatshirt, t-shirt, top and trousers. Catalog Track Pants were mapped to trousers; unsupported source categories were excluded.

2. Split and preprocessing

Split	Rows	Share
Training	10,795	69.9%
Validation	2,310	15.0%
Reserved test	2,336	15.1%

Identical SHA-256 image copies were grouped into the same split: **zero exact-image groups cross splits**. Training contained 4,931 t-shirts and 88 skirts, so moderate class weights were applied.

Images were EXIF-corrected, converted to RGB, padded to a square, resized to **224 x 224** and normalised with ImageNet statistics. Training used mild flips, rotation and colour jitter. A pixel cache avoided repeated decoding and resizing.

Training ran on a Kaggle Tesla T4. The reserved test split remained closed; the model comparisons below use validation data. Near-duplicate/product leakage remains an unresolved limitation.

Classification experiments

Transfer learning, model comparison, limited tuning and selection

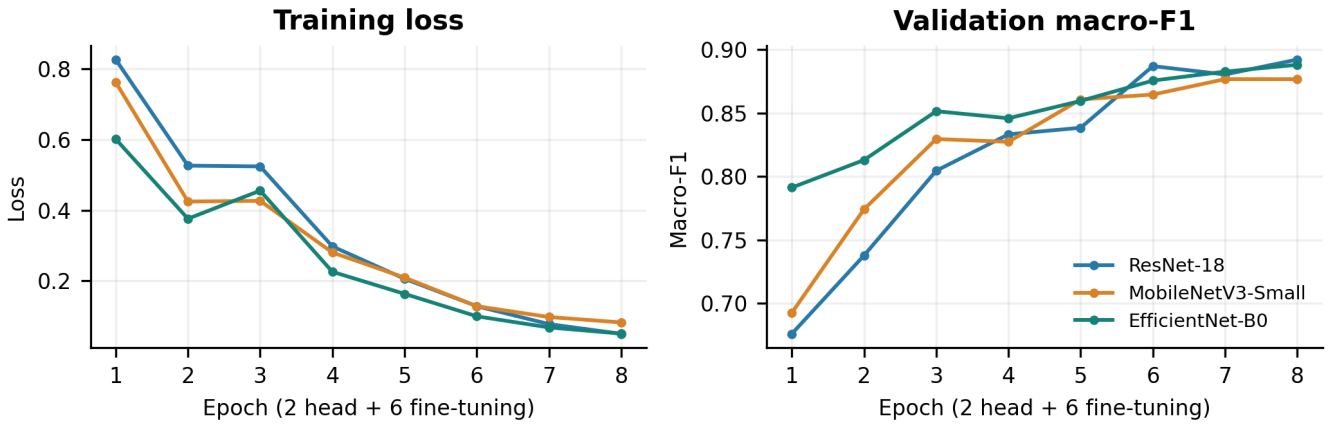
3. Training approach

Three ImageNet-pretrained models were compared using the same data and preprocessing. Each main experiment trained the new classification head for **2 epochs** with the backbone frozen, then fine-tuned the final blocks for up to **6 epochs**, using batches of 32. Best validation checkpoints were retained.

Model	Macro-F1	Accuracy	Top-2	Weights	CPU*
Majority: t-shirt	0.0572	45.93%	-	-	-
ResNet-18	0.8923	92.94%	98.87%	44.81 MB	30.64 ms
MobileNetV3-Small	0.8768	91.43%	98.61%	6.25 MB	10.94 ms
EfficientNet-B0	0.8882	93.16%	99.22%	16.39 MB	28.21 ms

All scores are measured on 2,310 validation images. Macro-F1 averages the F1 score of each class equally, so rare classes contribute as much as frequent ones. *CPU timings cover only the classifier forward pass on the Kaggle CPU.

Selected model: EfficientNet-B0. The selection rule chose the fastest CPU candidate within 0.01 macro-F1 of the best. EfficientNet reached **93.16% accuracy, 0.8882 macro-F1 and 99.22% top-2 accuracy**, with smaller weights than ResNet-18.



Recorded training loss and validation macro-F1 across the eight epochs of the main experiments.

4. Additional experiments and error analysis

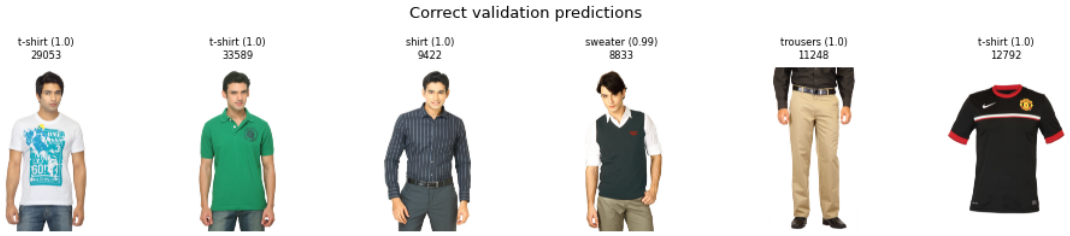
Two ResNet tuning trials were completed: longer fine-tuning obtained **0.8882 macro-F1**, and doubling the backbone learning rate obtained **0.8858**. Neither improved on the original 0.8923 result. A repeat with seed 43 reached **0.8757**, showing variation between runs. A third tuning trial was skipped.

The selected model made **158 validation errors**. T-shirt/top confusion accounted for 73. Lower F1 scores included sweatshirt (0.771), sweater (0.800) and top (0.802). Skirt F1 was 0.833, based on only 21 validation examples. Per-class metrics, confusion plots and error-image grids were saved.

Classification in practice

Saved validation examples: successful predictions and remaining mistakes

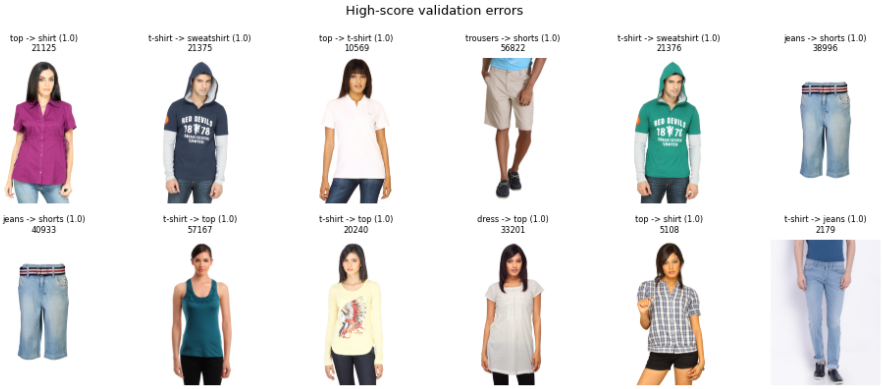
5. Examples of correct classification



Original notebook figure. From left to right: IDs 29053, 33589, 9422, 8833, 11248 and 12792. Scores are rounded in the figure; these six predictions agree with the mapped catalog category.

Product ID	Catalog category	Predicted category	Raw score
29053	t-shirt	t-shirt	1.000000
33589	t-shirt	t-shirt	0.999999
9422	shirt	shirt	0.999999
8833	sweater	sweater	0.985493
11248	trousers	trousers	0.999999
12792	t-shirt	t-shirt	0.999965

6. Examples of high-score mistakes



Original notebook error grid. Labels use catalog target -> model prediction; displayed scores are rounded. These examples were selected to illustrate failure modes, not to represent the error rate of all photos.

What the examples show. Visually close categories can be confused: top/shirt, t-shirt/sweatshirt and jeans/shorts. Multiple visible garments can also distract the classifier. Some catalog targets are debatable, so catalog disagreement alone cannot establish which visual label is correct.

Practical interpretation. A raw softmax score close to 1.0 does not guarantee correctness. The exported interface identifies scores as uncalibrated and requests user review. Across all validation images, 2,152 predictions agreed with the catalog and 158 did not.

Automatic descriptions

Attribute predictions, structured fields and deterministic text rendering

7. Description pipeline

Three small linear heads were trained on training-row catalog labels to predict **sleeve length, pattern and colour**. They reuse the selected classifier's features plus a colour histogram, normalised using training statistics. Missing and unsupported attribute labels were excluded from the relevant head evaluation.

Descriptions are generated by a **deterministic template**, rather than a generative language model. It renders supported structured fields and omits unknown values. Colour remains a suggestion until the user confirms it. Every prediction is flagged for user review, and corrections can override model fields.

Attribute	Validation result	Description policy
Sleeve length	92.74% agreement; macro-F1 0.8718; 1,639 usable labelled rows.	Stated for supported categories: jacket, shirt, sweatshirt, t-shirt and top.
Pattern	82.67% overall agreement on 1,835 labelled rows. At threshold 0.97: 72.53% coverage and 9.32% error.	Accepted above score 0.97 for supported categories. The word solid is omitted from the sentence.
Colour	69.05% agreement; macro-F1 0.6427; 2,291 usable labelled rows.	Two suggestions offered; enters the sentence after user confirmation.

These metrics measure agreement with weak catalog metadata, not human-verified factual correctness. Threshold exploration reused validation data.

8. Generated description results

Sentence outcome on all validation images	Count
Category + two attributes mentioned	998
Category + one attribute mentioned	831
Known category only	375
Category withheld: A garment.	106
Total	2,310

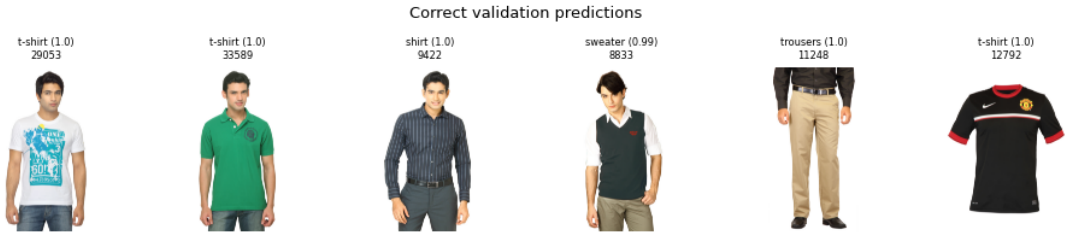
1,829 descriptions (79.2%) mention at least one attribute beyond category. Accepted structured values and words actually rendered are counted separately; for example, a solid pattern can be present in the fields without appearing in the sentence.

Examples recorded in the outputs:
Product 10065: A printed short-sleeved t-shirt.
Product 16392: A short-sleeved shirt.
Product 39386: A pair of jeans.

Description examples

From a garment photo to validated fields and a short readable sentence

9. Same photos, richer outputs



The photos above are the same six classification examples. The table below uses their saved description outputs; automatic colour is deliberately omitted.

Product ID	Pattern / sleeve fields	Rendered description
29053	printed / short	A printed short-sleeved t-shirt.
33589	unknown / short	A short-sleeved t-shirt.
9422	striped / long	A striped long-sleeved shirt.
8833	unknown / unknown	A sweater.
11248	solid / unknown	A pair of trousers.
12792	printed / short	A printed short-sleeved t-shirt.

10. A worked example: product 12792

Input: one garment photo. **Category prediction:** t-shirt, raw score 0.999965. **Accepted attributes:** printed pattern and short sleeves. The category comes from the classifier; the extra attributes come from the trained attribute heads.

```
{"category": "t-shirt", "color": "unknown",
"pattern": "printed", "sleeve_length": "short"}
```

Automatic sentence: A printed short-sleeved t-shirt.
User action: review the category and attributes, confirm or correct them, then save the approved wardrobe item. User-confirmed colour may be included after confirmation.

11. Examples of deliberate omission and errors

Product 11248: pattern is predicted as solid, but the renderer omits that word: A pair of trousers.
Product 8833: no supported extra attributes are accepted: A sweater.
Product 8587: output is A checked long-sleeved shirt. The catalog says Short Sleeves. This is an attribute disagreement requiring review, even though the category is correct.

Quality of the text versus quality of the prediction. Schema checks verify allowed values and valid output structure. They do not prove that each attribute matches the image. Unknown fields stay in the JSON and are omitted from the sentence; no fabric, brand, size or fit claims are generated.

Quality checks and export

Optional image-quality assistant and the final notebook outputs

12. Optional image-quality assistant

A rule-based module measures resolution, sharpness, brightness and contrast, and returns warnings with suggested actions. It requires no additional trained model. Readable-image warnings are advisory; unreadable inputs are reported separately.

The blur threshold was calibrated on **300 validation photos** and blurred copies. Evaluation used **60 of those photos**, each with four strong synthetic degradations. It therefore demonstrates rule behaviour on calibration-related examples, rather than independent phone-photo performance. The chosen sharpness threshold was **13.84**.

Image set	Images	Observed result
Unmodified catalog photos	60	0% returned any warning.
Strongly blurred	60	100% returned the expected blur warning.
Strongly darkened	60	100% returned the expected darkness warning.
Strongly brightened	60	100% returned the expected brightness warning.
Downscaled	60	100% returned the expected resolution warning.



Saved notebook example: an unmodified photo and its blurred, darkened, brightened and downscaled variants.

13. Exported artifacts and verification

selected_model_bundle.zip: 25 files, approximately 15.3 MB compressed. Includes classifier weights, attribute heads, inference and description code, class mapping, preprocessing, review policy, quality checks and checksums.
bo6_results.zip: approximately 238 MB, containing experiments, metrics, figures, candidate model exports and the selected bundle.

The core export passed **22/22 software checks** in a fresh process using 3 synthetic and 44 validation images. After adding quality checks, the portable model self-test passed again. These checks establish software consistency; real-world performance remains to be evaluated.

Remaining evaluation: reserved catalog test, independent DressMe phone photos and human review of description factuality. Near-duplicate/product leakage remains unresolved. Quality thresholds are provisional. The notebook delivers both mandatory ML components and an optional quality assistant; application integration is the next phase.